

Compiler Design

Phase 3

Semantic Analyzer

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Instructions

- All the problems relating to the Syntax Phase and Lexical Phase should be resolved first before starting this phase.
- The main objective of this phase is Type Checking.
- All the assignment expressions can be of the form

int = int char = char is correct

However *int = char char = int* are errors.

- The expressions like that of addition subtraction etc .. will resolve to

*Char + char $\rightarrow$ char int + int $\rightarrow$ int are all correct while
int+ char char + int are errors.*

- The expression having a relational operator will always resolve to an integer and this is possible only if both sides of the relational operator resolve to same type.i-e

if int < int or char < char then the type returned will be integer.

Other wise everything else will be treated as an error.

- The expressions in the For loop will be of integer type only that is the loop will be from integer to integer with an increment in integer .For e.g.

For (k = 2 ; j < 2 ; k =k+1) here the type of k , j is integer . and all these inner expressions resolve to integers only .

- The conditional expression in the while loop and if else will only be of integer type. This follows from the fact that the relational expressions will only resolve to integer type.

- Some errors that are reported in this phase are
 - Undeclared identifier.
 - Multiple declarations of same identifier
 - Type mismatch in statement or expression.
 - Integer expected in conditional expression.

----- **The End** -----